

Repository Guide: A Dynamic Scheduler

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Overview

This document provides a complete guide to the repository accompanying the thesis “*Cloudy with a Chance of Data: A Dynamic Scheduler for All-Sky Surveys with Real-Time Optimization*”.

All file paths are relative to the repository root.

1 Scripts (executable Python modules)

Table 1: Main scripts and their functions

File	Description
<code>scripts/dream_extract_server.py</code>	Core extraction: loads DREAM HDF5 from S3, projects to Cartesian grid, computes absolute minimum and motion-tracking pointings, and saves <code>.npz</code> files.
<code>scripts/dream_batch.py</code>	Batch wrapper around <code>dream_extract_server.py</code> for many nights.
<code>scripts/run_dream_scheduler_grid.py</code>	Full grid search: runs the greedy scheduler (Rubin OpSim) on each DREAM night, computes metrics, and saves comparisons.
<code>scripts/dream_night_classifier.py</code>	Classifies each night as CLEAR / PARTIALLY CLOUDY / UNIFORM THIN / FULLY CLOUDY based on mean and variance of cloud extinction.
<code>scripts/dream_cloud_classification.py</code>	Applies K-means clustering to partially cloudy nights to identify typical cloud patterns.

2 Jupyter Notebooks (root directory)

Table 2: Analysis and pipeline notebooks

File	Description
<code>m5.ipynb</code>	Complete pipeline: moon avoidance, seeing correction, m_5 depth calculations, ConsDB comparison, and slew-gated photon collection.
<code>metrics.ipynb</code>	MAF-style metric suite that generates CSV metrics and plots in <code>metrics/</code> and <code>metrics/maf_plots/</code> .
<code>tradeoff.ipynb</code>	Theoretical gain analysis exploring cloud, slew, and zenith weight trade-offs.
<code>moon.ipynb</code>	Moon illumination correction, zero-point analysis, and figure generation in <code>moon_plots/</code> .
<code>contrails.ipynb</code>	Hough transform detection of contrails and creation of avoidance zones.
<code>airplanes.ipynb</code>	ADS-B aircraft trajectory prediction for the Rubin field of view.
<code>corrections.ipynb</code>	Bias subtraction and systematics removal from DREAM data.
<code>final_plots.ipynb</code>	Assembles thesis-ready figures including gain boxplots and sky maps.
<code>figures.ipynb</code>	Additional figure generation for cloud motion and scheduler evolution.
<code>boundary_method.ipynb</code>	Older cloud boundary detection method retained for reference.
<code>binning.ipynb</code>	Tests spatial binning effects on cloud maps.
<code>combinatorix.ipynb</code>	Combinatorial analysis of scheduler and cloud-night pairings.
<code>crosscorr_pred.ipynb</code>	Cross-correlation prediction validation.
<code>stats.ipynb</code>	Statistical summaries of metric distributions.
<code>ml_method.ipynb</code>	Machine learning classification experiments (abandoned).
<code>tracking.ipynb</code>	Older motion tracking implementation retained for reference.

3 Data Directories

`dream_data/` — DREAM all-sky camera data

Table 3: DREAM data contents

File / subdirectory	Description
frames/	Individual <code>.npz</code> arrays representing single cloud extinction maps.
dream_20260126/	Batch of raw HDF5 files.
dream_20260126_manifest.csv	Index of HDF5 files with timestamps and S3 URLs.
dream_night_20250715.npz	Example processed night containing cloud grids and pointings.
dream_night_20m_stats*.csv	Statistics aggregated over 20-minute bins.

rubin_sim_data/ — Rubin Observatory simulation inputs

Table 4: Rubin simulation data

File / subdirectory	Description
site_models/	Atmospheric models including <code>cloud.db</code> , <code>seeing.db</code> , and related files.
pointings/	Pointing tensors and scheduler plans for nights 2161, 2210, and 2278.
pointing_tensor_all_nights.npz	Combined pointing tensor for all nights.
initial_example.ipynb	Tutorial notebook for Rubin Sim.

4 Output Directories

metrics/ — Per-night metric outputs

Table 5: Metrics directory structure

File / subdirectory	Description
metrics_data*.csv	Raw metric tables per night.
metrics_*.png / *.pdf	Single-night metric plots.
maf_results_summary.csv	Aggregated MAF metrics across all nights.
maf_sky_maps.png	Sky maps of coadded depth and visit distributions.
maf_scheduling_comparison.png	Comparison of scheduling strategies.
maf_plots/	Histograms for each metric.
broader_metrics/	Cross-comparison plots.

File / subdirectory	Description
maf_multinight_results/	Subdirectories for each processed night containing sky maps, metrics, and CSV outputs.

comp_plots/ — DREAM vs ConsDB comparisons

Table 6: Comparison plots

File	Description
dream_vs_consdb_zp.png	Zero-point comparison between DREAM and ConsDB.
dream_vs_consdb_extinction_*.png/pdf	Extinction comparisons in different observing bands and conditions.
dream_zps_vs_consdb_*.png	Per-night zero-point scatter plots.
dream_zeropoint_iqr_results.csv	IQR statistics of zero-point differences.
dream_zp_scatter_pairs.csv	Pairwise zero-point scatter data.

5 Data Flow Summary

```
DREAM HDF5 (S3)
->
scripts/dream_extract_server.py
->
dream_night_YYYYMMDD.npz

OpSim SQLite
->
scripts/run_dream_scheduler_grid.py
->
greedy pointings

m5.ipynb / metrics.ipynb
->
metrics/ + comp_plots/ + figures/
```

Reproducing the Thesis Results

All required input data (except large DREAM HDF5 files) are included in the repository. Raw DREAM data can be obtained through query Chronograf for HDF5 file paths.